ECON 0100 | Demo D

Demos are similar to Checkpoints, often taken directly from past semesters. Work through the problems and check your work against mine. Practice answering clearly and completely. You are encouraged to work in small groups. Find a study room, grab some classmates, and work together on a whiteboard.

Hermione's Potion Ingredients

Hermione and Draco share a garden plot growing moonflowers for their potions project. Each can harvest 1 or 2 flowers weekly. More harvesting reduces future growth. The payoff matrix shows their utility from different harvesting choices.

		Draco H1	**Draco** H2
Hermione	Harvest 1 (H1)	10, 10	6, 15
	Harvest 2 (H2)	14, 6	8, 8

Outcomes are written (Hermione's choice, Draco's choice).

D1.1 | Best Responses

	Hermione	**Draco**
Best Response When the Other Harvests 1:	_____	_____
Best Response When the Other Harvests 2:	_____	_____
Dominant Strategy:	☐ H1 ☐ H2 ☐ none	☐ H1 ☐ H2 ☐ none

D1.2 | Nash Equilibrium & Efficiency

Nash Equilibrium (*check all that apply*):	☐ (H1, H1)	☐ (H1, H2)	☐ (H2, H1)	☐ (H2, H2)
Socially Efficient Outcome (*check all that apply*):	☐ (H1, H1)	☐ (H1, H2)	☐ (H2, H1)	☐ (H2, H2)
Total Payoff, Nash Equilibrium / Efficient Outcome:	_____ / _____			
An Outcome Both Prefer to the Nash Equilibrium:	☐ (H1, H1)	☐ (H1, H2)	☐ (H2, H1)	☐ none

D2.1 | Common Resources

The Moonflowers in the Plot Are: ☐ rival ☐ non-rival ☐ excludable ☐ non-excludable

If Hermione Switches From (H1, H2) to (H2, H2): Hermione: _____ Draco: _____

Professor Sprout begins checking the plot each week and fines anyone who harvests 2 flowers. The fine lowers that student's payoff by 6. Fill in the payoff matrix with the fine.

Nash Equilibrium With the Fine:
☐ (H1, H1) ☐ (H1, H2)
☐ (H2, H1) ☐ (H2, H2)

		Draco H1	**Draco** H2
Hermione	H1		
	H2		

End-of-Term Fireworks

Five prefects are planning the end-of-term celebration. A fireworks show over the lake would cost 60 coins, and it can be seen from anywhere on the grounds. The table shows each prefect's benefit from the celebration (in coins), both without and with the fireworks.

Benefits (*in coins*)	**Hermione**	**Draco**	**Hannah**	**Cho**	**Padma**
Without fireworks	25	22	18	16	12
With fireworks	40	35	32	20	15

D3.1 | Public Goods

The prefects first try to raise the 60 coins by splitting the cost into voluntary contributions of 12 coins. Each prefect decides in their own interest whether to contribute or not.

The Fireworks Show Is: ☐ rival ☐ non-rival
☐ excludable ☐ non-excludable

Prefects Who Would Contribute Voluntarily: _____

	Hermione	**Draco**	**Hannah**	**Cho**	**Padma**
Marginal benefit of the fireworks					

Marginal Social Benefit of the Fireworks: _____
Providing the Fireworks Is: ☐ efficient ☐ not efficient

D3.2 | Voting on Public Goods

Instead, the prefects put the fireworks to a vote. If a majority votes for them, each prefect pays a fee of 12 coins (60 coins total).

Hermione's Benefit Without Fireworks: _____
Hermione's Benefit With Fireworks (*after the fee*): _____
Hermione's Vote: ☐ for ☐ against
Prefects Voting For / Against: _____ / _____
Total Benefit Without Fireworks: _____
Total Benefit With Fireworks (*after the fees*): _____
Fireworks With a 12-Coin Fee: ☐ efficient ☐ not efficient